



CSE 471: Machine Learning

Decision Tree

Decision Tree

Decision tree learning is a method for approximating discrete valued target functions, in which the learned function is represented by a decision tree. Decision tree learning is one of the most widely used and practical methods for inductive inference.

Decision tree classifiers are attractive models if we care about interpretability. As the name "decision tree" suggests, we can think of this model as breaking down our data by making a decision based on asking a series of questions.



Two types of decision trees

There are two types of decision trees.

1. **Classification trees**

Tree models where the target variable can take a discrete set of values are called *classification trees*. In these tree structures, leaves represent class labels and branches represent conjunctions of features that lead to those class labels.

2. **Regression trees**

Decision trees where the target variable can take continuous values (real numbers) like the price of a house, or a patient's length of stay in a hospital, are called *regression trees*.

Some well-known decision tree algorithms

1. ID3 (Iterative Dichotomiser 3) developed by Ross Quinlan
2. C4.5 developed by Ross Quinlan
3. C5.0 developed by Ross Quinlan
4. CART (Classification And Regression Trees)
5. 1R (One Rule) developed by Robert Holte in 1993.
6. RIPPER (Repeated Incremental Pruning to Produce Error Reduction)
Introduced in 1995 by William W. Cohen.

As an example of decision tree algorithms, we discuss the details of the ID3 algorithm and illustrate it with an example.

The ID3 algorithm

Ross Quinlan, while working at University of Sydney, developed the ID3 (Iterative Dichotomiser 3) algorithm and published it in 1975.

Assumptions

- The algorithm uses information gain to select the most useful attribute for classification.
- We assume that there are only two class labels, namely, “+” and “-”. The examples with class labels “+” are called positive examples and others negative examples.

The algorithm

Notations

The following notations are used in the algorithm:

S	The set of examples
C	The set of class labels
F	The set of features
A	An arbitrary feature (attribute)
$\text{Values}(A)$	The set of values of the feature A
v	An arbitrary value of A
S_v	The set of examples with $A = v$
Root	The root node of a tree

Algorithm ID3(S, F, C)

1. Create a root node for the tree.
2. **if** (all examples in S are positive) **then**
3. **return** single node tree Root with label “+”
4. **end if**
5. **if** (all examples are negative) **then**
6. **return** single node tree Root with label “-”
7. **end if**
8. **if** (number of features is 0) **then**
9. **return** single node tree Root with label equal to the most common class label.
10. **else**
11. Let A be the feature in F with the highest information gain.
12. Assign A to the Root node in decision tree.
13. **for all** (values v of A) **do**
14. Add a new tree branch below Root corresponding to v .
15. **if** (S_v is empty) **then**
16. Below this branch add a leaf node with label equal to the most common class label in the set S .
17. **else**
18. Below this branch add the subtree formed by applying the same algorithm ID3 with the values $\text{ID3}(S_v, C, F - \{A\})$.
19. **end if**
20. **end for**
21. **end if**

Example

= DECISION TREE =

With ID3 (Iterative Dichotomiser 3)

⇒ Formulate the decision tree for the following Dataset.

Weather/outlook	Temp	Humidity	Wind	Play
sunny	Hot	High	Weak	No
sunny	Hot	High	Strong	No
cloudy	Hot	High	Weak	Yes
Rain	Mild	High	Weak	Yes
Rain	Cool	Normal	Weak	Yes
Rain	cool	Normal	Strong	No
cloudy	cool	Normal	Strong	Yes
sunny	Mild	High	Weak	Yes NO
sunny	cool	Normal	Weak	Yes
Rain	Mild	Normal	Weak	Yes
sunny	Mild	Normal	Strong	Yes
cloudy	Mild	High	Strong	Yes
cloudy	Hot	Normal	Weak	Yes
Rain	Mild	High	Strong	No

Example

Solution:

First of all, we need to identify the root node of the decision tree.

To identify the root node, we have to calculate the entropy and information gain of each feature.

Step 1: Entropy of entire dataset

$$S(+9, -5) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.94$$

Step 2: Information gain (IG) of weather

$$\bullet \text{ Entropy of sunny } (+2, -3) = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.92$$

$$\bullet \text{ Entropy of cloudy } (+4, -0) = -\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} \log_2 \frac{0}{4} = 0$$

$$\bullet \text{ Entropy of Rain } (+3, -2) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.92$$

$$\begin{aligned} \therefore \text{Information Gain} &= \text{Entropy Entire dataset} - \frac{5}{14} \text{Ent}(s) \\ &\quad - \frac{4}{14} \text{Ent}(c) - \frac{5}{14} \text{Ent}(R) \\ &= 0.242 \end{aligned}$$

Example

Step 3: IG of Temperature

$$\bullet \text{ Entropy of Hot } \{+2, -2\} = -\frac{2}{4} \log_2 \frac{2}{4} - \frac{2}{4} \log_2 \frac{2}{4} = 1$$

$$\bullet \text{ Entropy of Mild } \{+4, -2\} = -\frac{4}{6} \log_2 \frac{4}{6} - \frac{2}{6} \log_2 \frac{2}{6} = 0.92$$

$$\bullet \text{ Entropy of cold } \{+3, -1\} = -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} = 0.81$$

$$\begin{aligned} \therefore \text{Information Gain} &= \text{Entropy Entire dataset} - \frac{4}{14} \text{Ent}(H) - \frac{6}{14} \text{Ent}(M) \\ &\quad - \frac{4}{14} \text{Ent}(C) \\ &= 0.29 \end{aligned}$$

Step 4: IG of Humidity

$$\bullet \text{ Entropy of High } \{+3, -4\} = -\frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} = 0.98$$

$$\bullet \text{ Entropy of Normal } \{+6, -1\} = -\frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} = 0.59$$

$$\begin{aligned} \therefore \text{Information Gain} &= \text{Entropy Entire Dataset} - \frac{7}{14} \text{Ent}(H) - \frac{7}{14} \text{Ent}(N) \\ &= 0.15 \end{aligned}$$

Example

Step 5: Information Gain of Wind

$$\cdot \text{Entropy of strong } \{+3, -3\} = \frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1.0$$

$$\cdot \text{Entropy of weak } \{+6, -2\} = \frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8} = 0.81$$

$$\therefore \text{Information Gain} = \text{Entropy Entire Dataset} - \frac{6}{14} \text{Ent}(S) - \frac{8}{14} \text{Ent}(W) = 0.0428$$

Now, we find the entropy of all features as follows:

$$\text{I.G. Weather} = 0.242$$

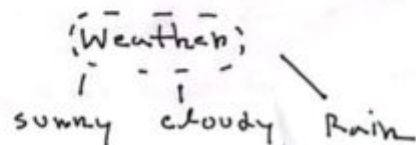
$$\text{I.G. Temp} = 0.029$$

$$\text{I.G. Humidity} = 0.15$$

$$\text{I.G. Wind} = 0.0428$$

Hence, Weather is the root node since it has the Highest I.G.

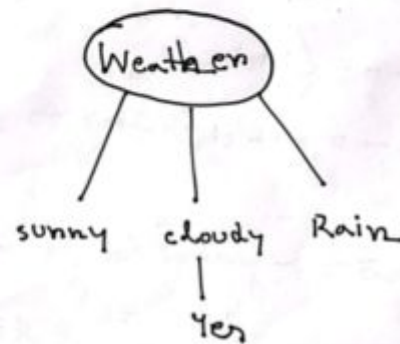
Therefore, the current tree is as follows:



Example

Now, we notice from the dataset that, whenever the weather/outlook is cloudy, the decision is always "Yes".

So, we can modify the tree as follows:



Now, we need to find the next node. If we look at the dataset, we have 5 sunny. So, we divide the dataset as follows:

Weather	Temp	Humidity	Wind	Play
sunny	Hot	High	Weak	No
sunny	Hot	High	strong	No
sunny	Mild	High	Weak	No
sunny	cool	Normal	Weak	Yes
sunny	Mild	Normal	strong	Yes

To decide the next node, we have to identify the IG of each feature from this dataset.

Example

2) step 6: Entropy of sunny $\{+2, -3\} = -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.92$

step 7: IG of Temp (for sunny)

• Entropy of Hot $\{+0, -2\} = -\frac{0}{2} \log_2 \frac{0}{2} - \frac{2}{2} \log_2 \frac{2}{2} = 0$

• Entropy of Mild $\{+1, -1\} = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1.0$

• Entropy of cool $\{+1, -0\} = -\frac{1}{1} \log_2 \frac{1}{1} - \frac{0}{1} \log_2 \frac{0}{1} = 0$

$$\therefore IG = \text{Entropy of sunny} - \frac{2}{5} \text{Ent}(H) - \frac{2}{5} \text{Ent}(M) - \frac{1}{5} \text{Ent}(C) \\ = 0.52$$

step 8: IG of Humidity (for sunny)

• Entropy of High $\{+0, -3\} = -\frac{0}{3} \log_2 \frac{0}{3} - \frac{3}{3} \log_2 \frac{3}{3} = 0$

• Entropy of Normal $\{+2, -0\} = -\frac{2}{2} \log_2 \frac{2}{2} - \frac{0}{2} \log_2 \frac{0}{2} = 0$

$$\therefore IG = \text{Entropy of sunny} - \frac{3}{5} \text{Ent}(H) - \frac{2}{5} \text{Ent}(N) \\ = 0.92$$

Example

Step 9: IG of Wind (For sunny)

$$\bullet \text{ Entropy of strong } \left\{ \begin{matrix} +1 \\ -1 \end{matrix} \right\} = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1$$

$$\bullet \text{ Entropy of Weak } \left\{ \begin{matrix} +1 \\ -2 \end{matrix} \right\} = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} = 0.919$$

$$\therefore \text{IG} = \text{Entropy of sunny} - \frac{2}{5} \text{Ent}(s) - \frac{3}{5} \text{Ent}(w) \\ = 0.013$$

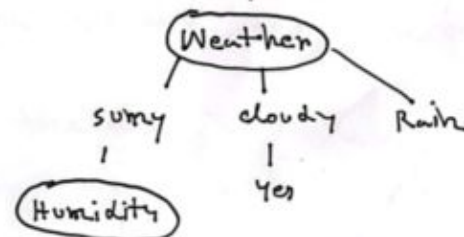
Now, we have

$$\text{IG of Temp for sunny} = 0.52$$

$$\text{IG of Humidity for sunny} = 0.92$$

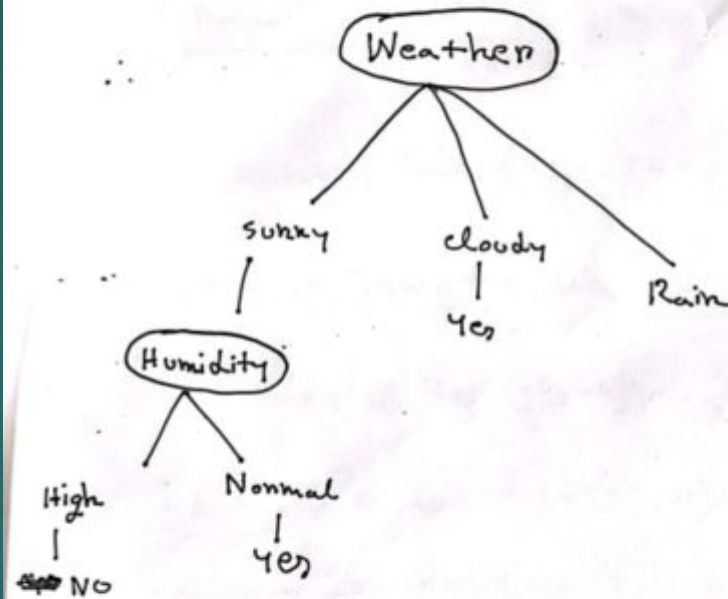
$$\text{IG of Wind for sunny} = 0.019$$

Hence, the Highest IG for sunny is Humidity. Therefore, we can update the ~~the~~ TREE as follows:



Example

If we observe the sunny dataset, whenever Humidity is High, decision is always "No", and vice-versa. So, we can update the ~~dataset~~ tree as follows:



We perform all the steps again for Rain. First We need to split the dataset for Rain.

Example

Weather	Temp	Humidity	Wind	Play
Rain	Mild	High	Weak	Yes
Rain	cool	Normal	Weak	Yes
Rain	cool	Normal	strong	No
Rain	Mild	Normal	Weak	Yes
Rain	Mild	High	strong	No

Step 10: Entropy Rain $\{+3, -2\} = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.92$

Step 11: IG of Temp for Rain

• Entropy of Hot $\{+0, -0\} = -\frac{0}{2} \log_2 \frac{0}{2} - \frac{0}{2} \log_2 \frac{0}{2} = 0$

• Entropy of ~~Hot~~ Mild $\{+2, -1\} = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.918$

• Entropy of cool $\{+1, -1\} = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1.0$

$$\begin{aligned}
 \therefore \text{IG of Temp} &= \text{Entropy of Rain} - \frac{0}{5} \text{Ent(H)} - \frac{2}{5} \text{Ent(M)} \\
 &\quad - \frac{2}{5} \text{Ent(c)} \\
 &= 0.019
 \end{aligned}$$

Example

Step 12: IG of Humidity for Rain

$$\bullet \text{ Entropy of High } \{+1, -1\} = -\frac{1}{2} \log_2 \frac{1}{2} - \frac{1}{2} \log_2 \frac{1}{2} = 1$$

$$\bullet \text{ Entropy of Normal } \{+2, -1\} = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.917$$

$$\therefore \text{IG} = \text{Entropy of Rain} - \frac{2}{5} \text{Ent}(S) - \frac{3}{5} \text{Ent}(W) \\ = 0.019$$

Step 13: IG of Wind for Rain

$$\bullet \text{ Entropy of strong } \{+0, -2\} = -\frac{0}{2} \log_2 \frac{0}{2} - \frac{2}{2} \log_2 \frac{2}{2} = 0$$

$$\bullet \text{ Entropy of weak } \{+3, 0\} = -\frac{3}{3} \log_2 \frac{3}{3} - \frac{0}{3} \log_2 \frac{0}{3} = 0$$

$$\therefore \text{IG} = \text{Entropy of Rain} - \frac{2}{5} \text{Ent}(S) - \frac{3}{5} \text{Ent}(W) \\ = 0.98$$

or, we have

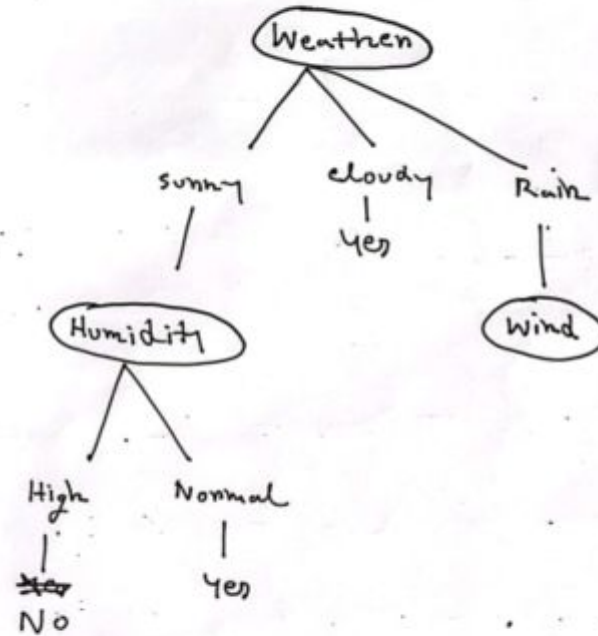
$$\text{IG of Temp for Rain} = 0.019$$

$$\text{IG of Humidity for Rain} = 0.019$$

$$\text{IG of Wind for Rain} = 0.98$$

Example

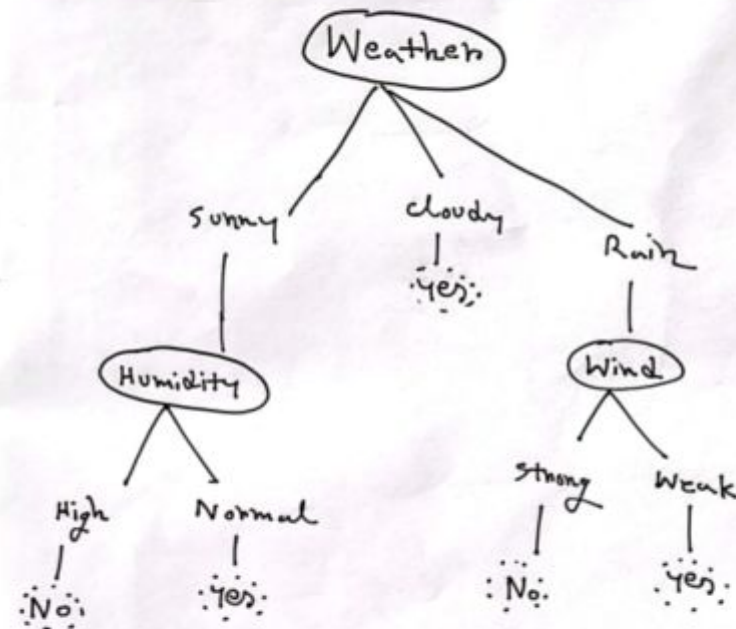
Here, the Highest IG for Rain is Wind. Therefore, we can update the tree as follows:



If we observe the Rain dataset, we see that whenever Wind is strong, decision is always "No" and vice-versa.

Example

Therefore, the final decision ~~the~~ tree is shown as below:



Now, we are able to make prediction based on this decision ~~the~~ ~~tree~~ TREE.



Some remarks

1. On the elements of a classification tree

The various elements in a classification tree are identified as follows.

- Nodes in the classification tree are identified by the feature names of the given data.
- Branches in the tree are identified by the values of features.
- The leaf nodes identified by are the class labels



Some remarks

2. On the order in which the features are selected

In the example discussed above, initially we chose the feature “sunny” to split the data set into disjoint subsets and then the feature “rain”, and so on. There was no theoretical justification for this choice. We could as well have chosen any other feature, as the initial feature for splitting the data. The classification tree depends on the order in which the features are selected for partitioning the data.

Some remarks

3. Stopping criteria

A real-world data will contain much more example record than the example we considered earlier. In general, there will be a large number of features each feature having several possible values. Thus, the corresponding classification trees will naturally be more complex. In such cases, it may not be advisable to construct all branches and leaf nodes of the tree. The following are some of commonly used criteria for stopping the construction of further nodes and branches.

- All (or nearly all) of the examples at the node have the same class.
- There are no remaining features to distinguish among the examples.
- The tree has grown to a predefined size limit.

Thank you!